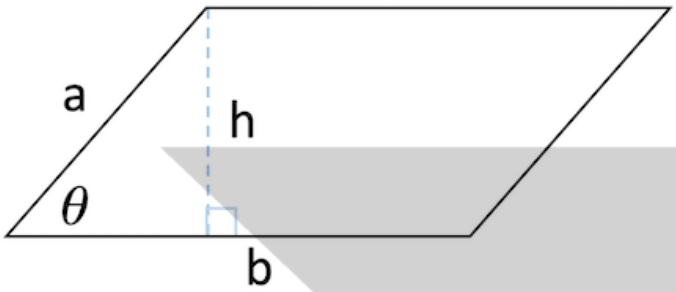
1. Mensuration

1.1. Area and Perimeter

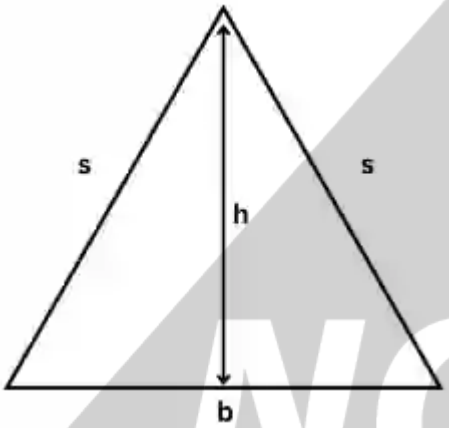
Parallelogram

- Area $= b \times h$ or $ab\sin(\theta)$
- Perimeter $= 2a + 2b$



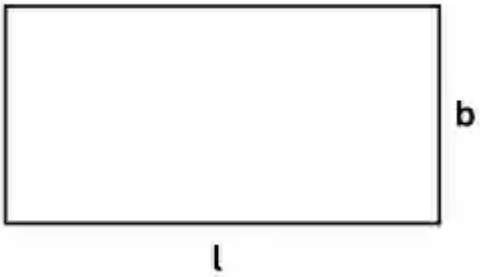
Triangle

- Area $= \frac{1}{2}b \times h$
- Perimeter $= b + 2s$
  - b = base , s = slant height



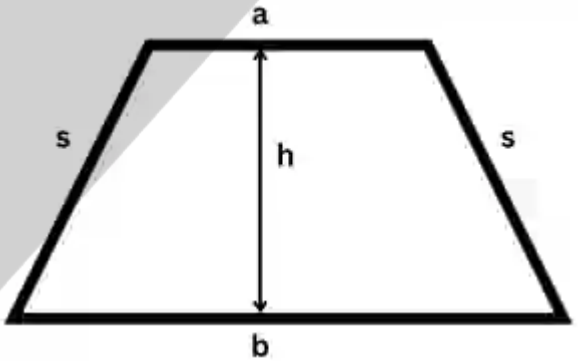
Rectangle

- Area $= l \times b$
- Perimeter $= 2l + 2b$
  - l= length , b= breadth



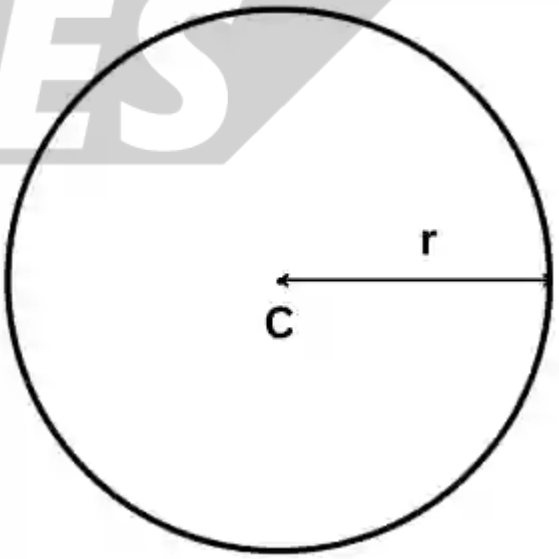
Trapezium

- Area $= \frac{1}{2}(a + b)h$
- Perimeter $= a + b + 2s$
  - s = slant height



Circle

- Area $= \pi r^2$
- Circumference $= 2\pi r$



Sector

- Area of Sector $= \frac{\theta}{360°} \times \pi r^2$

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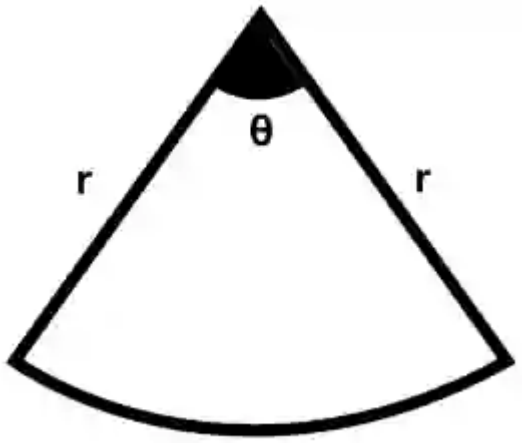
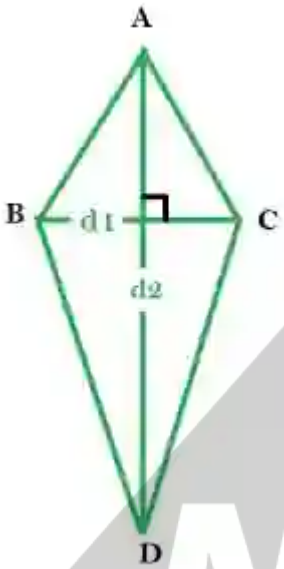
- Arc Length $= \frac{\theta}{360} \times 2\pi r$



Kite

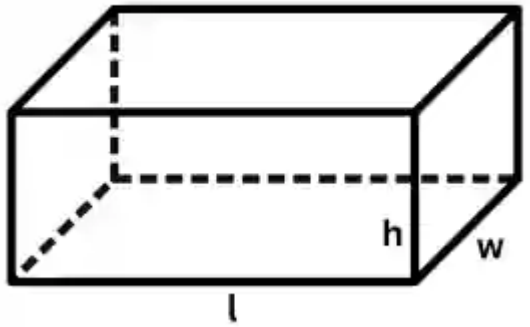
- Area $= \frac{d1 \times d2}{2}$
- Perimeter $= AB + BD + DC + CA$



1.2. Volume and Surface Area

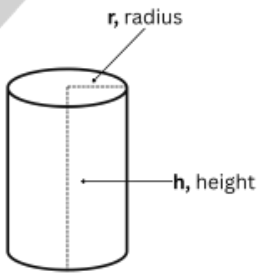
Cuboid

- Volume $= lwh$
- Surface Area $= 2(lw + wh + hl)$



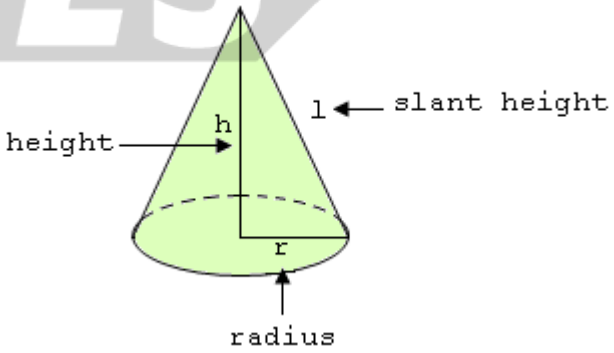
Cylinder

- Volume $= \pi r^2 h$
- Curved Surface Area $= 2\pi rh$
- Total Surface Area $= 2\pi r^2 + 2\pi rh$



Cone

- Volume $= \frac{1}{3}(\pi r^2 h)$
- Curved Surface Area $= \pi rl$
- Total surface Area $= \pi r^2 + \pi rl$
-



Sphere

- Volume $= \frac{4}{3}\pi r^3$
- Surface Area $= 4\pi r^2$

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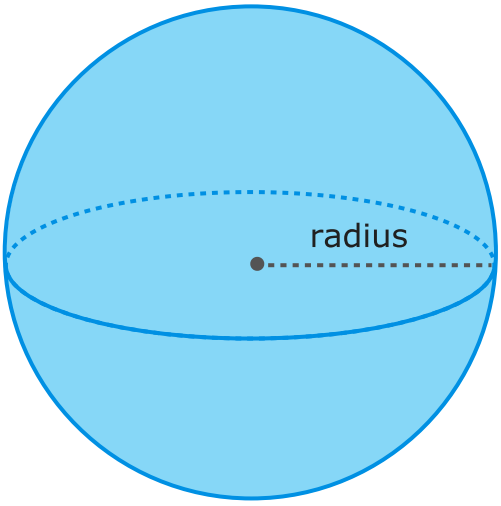
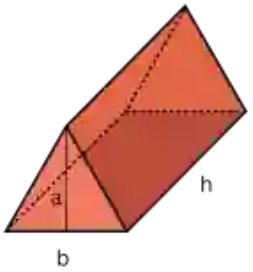
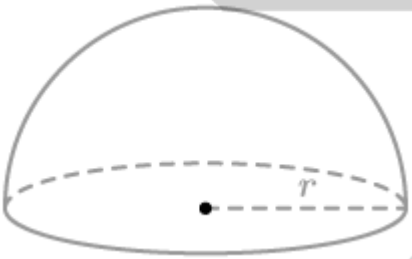
- 

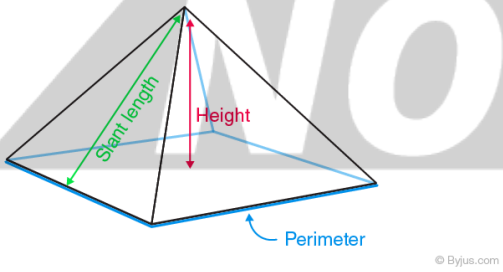


Hemisphere

- Volume $= \frac{2}{3}\pi r^3$
- Curved Surface Area $= 2\pi r^2$
- Surface Area $= 3\pi r^2$
- 

Pyramid

- Volume $= \frac{1}{3}b^2h$
- Surface Area $= b^2 + 2bs$
  - b = base length
  - s= slant height



Prism

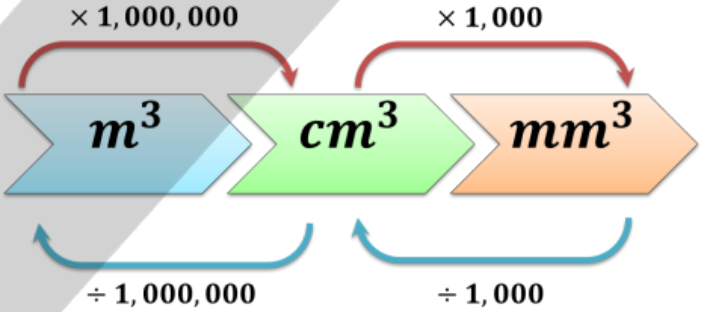
Volume $= \frac{1}{2}bah$

Curved Surface Area $= 2(b + h) \times a$

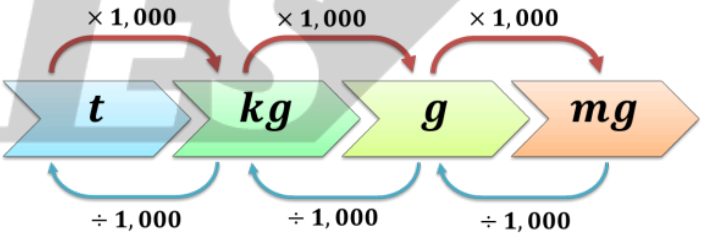
Total Surface Area $=$ Curved Surface Area $+ 2(b \times h)$

1.3. Units

Volume



Mass



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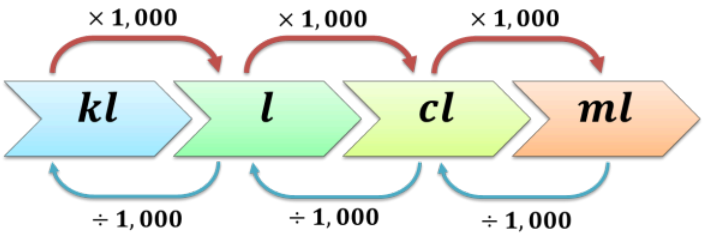
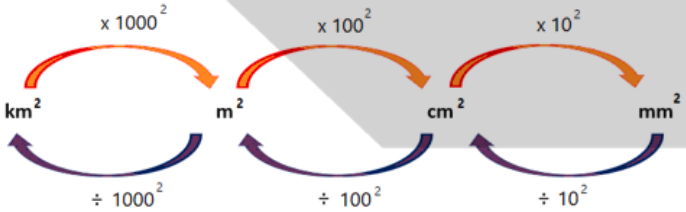
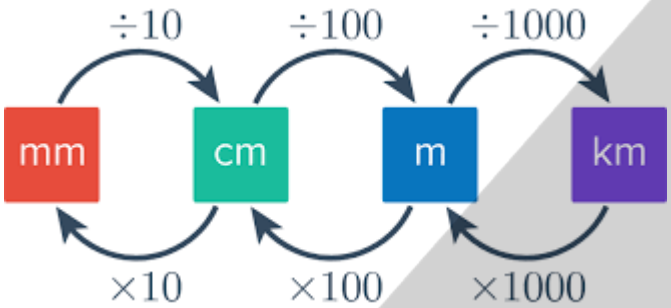
Capacity



Area



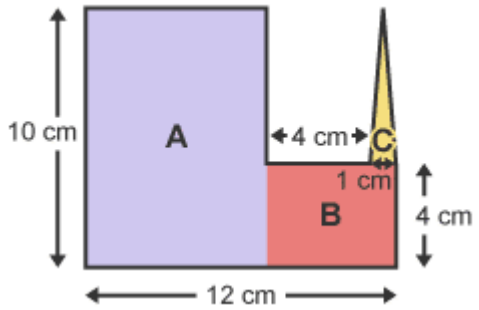
Length



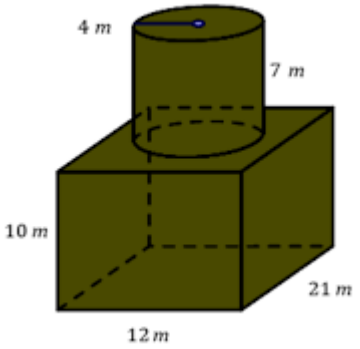
- Connecting Volume and Capacity:
  - $1ml = 1cm^3$
- Density $= \frac{\text{Mass}}{\text{Volume}}$

1.4. Compound Shapes and Parts of Shapes

- How to find Area and Perimeters of Compound Shapes
- 2D Shapes



- 3D Shapes



Surface Area of Shape:

Step 1: Surface Area of Cuboid (without the top face)

$2(lh + wh) + lw$

$= 2(12 \times 10 + 21 \times 10) + (12 \times 21)$

$= 2(120 + 210) + 252$

$= 2(330) + 252 = 660 + 252 = 912m^2$

Step 2: Remove circular hole (base of cylinder) from cuboid top

$\pi r^2 = \pi(4^2) = 16\pi m^2$

Adjusted cuboid surface area:

$912 - 16\pi$

Step 3: Curved Surface Area of Cylinder

$2\pi rh = 2\pi(4)(7) = 56\pi m^2$

Step 4: Add top circle of cylinder

$\pi r^2 = 16\pi m^2$

Step 5: Total Surface Area

$(912 - 16\pi) + 56\pi + 16\pi$

$= 912 + 56\pi m^2$

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